

Program: Diploma in Civil Engineering and Planning	
Course Code: 4398	Course Title: Estimation and Valuation Lab
Semester: 4	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To impart the knowledge about the procedure for estimating and costing of Civil Engineering works
- To develop in the student the skill to use spreadsheet for the preparation of detailed estimate and abstract of estimate for construction projects.
- To perform rate analysis for different items associated with construction projects.
- To prepare valuation certificates and statements of a real and a landed property.

Course Prerequisites:

Topic	Course code	Course name	Semester
Elementary Mathematics		Mathematics I and II	1 & 2
Building Drawing and Planning		Building Drawing and Planning	3
Building construction and construction materials		Building construction and construction materials	3

Course Outcomes:

CO n	Description	Duration (Hours)	Cognitive level
CO1	Identify the basic principles for estimation and costing of Civil Engineering works	9	Understanding
CO2	Prepare detailed estimate for infrastructure works using spreadsheet	14	Applying
CO3	Prepare abstract of estimate for construction projects using spreadsheet.	12	Applying
CO4	Prepare detailed estimate, abstract of estimate and valuation certificates of a building	6	Applying
	Series Tests	4	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3		3				
CO3		3					
CO4	3				3		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

On completion of the course student will be able to:

Module Outcomes	Description	Duration (Hours)	Cognitive level
CO1	Identify the basic principles for estimation and costing of Civil Engineering works		
M1.01	Prepare the list of items to be executed with units for detailed estimate of a given structure from the given drawing.	3	Understanding
M1.02	Prepare a report on market rates for given material, labour wages, hire charges of tools & Equipments required to construct a Civil Engineering Structure.	3	Understanding
M1.03	Study of items with specification given in DSR using PRICE Software. (any ten items)	3	Understanding
CO2	Prepare detailed estimate for infrastructure works using spreadsheet		
M2.01	Calculate the quantity of items of work from the given set of drawings using standard measurement sheet for load bearing residential structure using description of item from DSR (1BHK Building with staircase).	3	Applying
M2.02	Prepare detailed estimate of road of one kilometer length from the given drawing. (CC Road and WMM)	3	Applying
M2.03	Prepare detailed estimate of well from the given set of drawing.	2	Applying

M2.04	Prepare detailed estimate of small Septic tank and soak pit from the given set of drawings.	3	Applying
M2.05	Prepare detailed estimate of a steel roof truss	3	Applying
	Series Test-1	2	

CO3	Prepare abstract of estimate for construction projects using spreadsheet		
M3.01	Calculate the reinforcement quantities from the given set of drawings for a room size of 3 m x 4 m with bar bending schedule (footing, column, beam, lintel with chajja, slab)	4	Applying
M3.02	Prepare bill of quantities of given item from actual measurements (any four items)	4	Applying
M3.03	Prepare rate analysis and record in Measurement Book (MB) for any four items.	4	Applying
CO4	Prepare detailed estimate, abstract of estimate and valuation certificates of a building		
M4.01	Open ended experiment- Detailed estimate and abstract of estimate, valuation certificate of a building.	6	Applying
	Series Test II	2	

Text / Reference:

T/R	Book Title/Author
T1	Datta, B.N., Estimating and Costing in Civil engineering, UBS Publishers Distributors Pvt. Ltd. New Delhi.
R2	Rangwala, S.C., Estimating and Costing, Charotar Publishing House PVT. LTD., Anand.
R3	Birdie, G.S., Estimating and Costing, Dhanpat Rai Publishing Company(P) Ltd. New Delhi.
R4	V N Vazirani & S P Chandola, Civil engineering Estimating and Costing, Khanna Publishers
R5	Chakraborti, M., Estimating and costing, specification and valuation in civil engineering, Monojit Chakraborti, Kolkata.
R6	IS 1200-1968; Methods of measurement of Building & Civil Engineering works.
R7	CPWD Schedule of Rates for Delhi
R8	CPWD Analysis of Rates for Delhi- Vol I and II

R9	Ministry of Road Transport and Highways (MORT&H) Specifications and Analysis of Schedule of Rates.
R10	IRC SP: 20 Rural Roads Manual

Online Resources:

Sl.No	Website Link
1	https://nptel.ac.in/
2	http://egyankosh.ac.in/
3	https://swayam.gov.in/
4	http://www.vlab.co.in/